



FIRST SEMESTER, 2026–2027
Course Handout Part II

Date: 2nd Aug. 2026

In addition to part I (General Handout for all courses appended to the Timetable), this portion gives further specific details regarding the course.

Course No : EEE F472
Course Title : Satellite Communications
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1. Course Description:

Review of microwave communications and LOS systems; the various satellite orbits like GEO, MEO, LEO; the satellite link analysis and design; the communication transponder system like INSAT, INELSAT etc; the earth segment and earth station engineering; the transmission of analog and digital signals through satellite and various modulation techniques employed; the multiple access techniques like FDMA, TDMA, CDMA, DAMA, etc; the INSAT program; salient features of INSAT – systems and services offered; satellite services offered by INTELSAT, INMARSAT and future satellites like IRIDIUM etc; future trends in satellite communications.

2. Scope & Objectives of the Course:

This course introduces satellite communication systems, combining radio-wave propagation, orbital mechanics, satellite subsystems, modulation, and coding. Satellite link analysis and link-budget design form the core, applied hands-on component of the course, reinforced through in-class problem-solving sessions and selected simulation tasks. Students then study fixed-assignment multiple-access techniques (FDMA/TDMA/CDMA) and briefly explore random-access schemes, followed by applications in VSAT, mobile satellite, and GPS/DGPS. By the end of the course, students will be able to analyze satellite communication systems, able to design satellite links and evaluate their performance, develop and defend a link-budget analysis, and determine satellite position/coverage for a given orbit.

3. Course learning outcomes

Some of the key learning outcomes are stated below.

- CLO1.** Explain the fundamental concepts, historical evolution, segments, and subsystems of satellite communication systems.
- CLO2.** Analyze radio-wave propagation effects and orbital mechanics (LEO/MEO/GEO) and their influence on system and link design.
- CLO3.** Design and evaluate satellite RF uplink/downlink link budgets, including noise and intermodulation considerations, using hand calculations and simulation tools.

- CLO4.** Apply and compare digital modulation, coding, and multiple-access techniques (FDMA/TDMA/CDMA) for satellite systems.
- CLO5.** Identify and evaluate VSAT, mobile satellite links, understand GPS/GNSS system architectures and their performance limitations
- CLO6.** To understand, appreciate and analyze recent trends in satellite communications such as next generation non-terrestrial network/space-air-ground integrated network.

4. Textbooks:

- TB1.** *Satellite Communication Systems*, Pratt, Bostian, and Allnutt, 2nd Edition, John Wiley & Sons, 2019.

References:

- Ref1.** *Satellite Communications*, Robert M. Gagliardi, 2nd Edition, Springer, 2012.
- Ref2.** *Satellite Communications*, Dennis Roddy, McGraw-Hill Professional, 2001.
- Ref3.** *Satellite Communications Systems*, G. Maral and M. Bousquet, 5th Edition, John Wiley & Sons, 2009.
- Ref4.** *Digital Satellite Communication*, Tri T. Ha, 2nd Edition, MGH.

5. Course Plan:

This undergraduate-level satellite communication course deals with the fundamentals of satellite communications. The course comprises six modules: M1) Module 1 consists of an introduction, historical developments, and research trends. M2) Module 2 covers satellite orbits, orbit elements, orbital effects, and Friendship7 spacecraft. M3) Module 3 comprises satellite link parameters, uplink, and downlink budget. M4) The fourth module consists of modulation, multiplexing, and error control coding. M5) The fifth module covers multiple access techniques. M6) The last module includes selected topics such as very small aperture terminal (VSAT), mobile sat (MSAT), the global positioning system (GPS), differential GPS (DGPS) and recent trends and advances.

Module	Description	Reference	CLO
M1	L1–L2: Introduction, motivation, history, TRL, satellite segments & definitions. L3: Classify sample missions by TRL/segment	Ch.1–TB	CLO1
M1	L4–L5: Radio-wave propagation and frequency-selection effects. L6: Rain-fade/attenuation calculation	Ch.8–TB	CLO1, CLO2
M2	L7–L8: Orbits–LEO/MEO/GEO, merits/demerits, launch vehicles. L9: Orbital mechanics related concepts (selected)	Ch.2,10,11–TB/RB1	CLO2

Continued on next page

Module	Description	Reference	CLO
M2	L10: Segments, space environment/physical medium. L11–L12: Spacecraft subsystems (power, telemetry, AOCS, sensors, thermal, propulsion).	Ch.3–TB, RB2/RB3	CLO1
M2	L13: Subsystems continued. L14–L15: Communication payload/transponder (antenna, LNA, HPA–SSPA/TWTA). Transponder block-diagram design.	Ch.3–TB, Ch.9–RB1	CLO1
M2/M3	L16: Reliability in satellite system design. L17–L18: RF link analysis intro–EIRP, G/T.	Ch.3–TB, Ch.13–RB4, Ch.4–TB	CLO1, CLO3
M3	L19–L21: Uplink/downlink C/N, C/N ₀ , overall link budget. Preliminary link-budget design and examples	Ch.4–TB, Ch.5–RB1	CLO3
M3	L22–L23: Intermodulation (IM), two-tone third-order IM. L24: IM noise concept, illustration, examples.	Ch.6–TB	CLO3
M4	L25–L27: Analog & digital signals, modulation and coding schemes. Constellations and BER (reliability measure).	Ch.5,7–TB, RB2/RB3	CLO4
M4/M5	L28: Modulation/coding wrap-up. L29–L30: Multiple access–FDMA, TDMA.	Ch.6–TB, Ch.6–RB1	CLO4
M5	L31–L32: CDMA; on-board/regenerative processing. L33: Capacity-comparison problems for FDMA/TDMA/CDMA. Review and other techniques	Ch.6–TB, RB1	CLO4
M6	L34–L35: VSAT system planning & earth-station engineering. L36: VSAT network sizing, dependent parameters.	Ch.9–TB	CLO5
M6	L37–L38: Mobile satellite communication, non-GEO systems, fading & remedies. L39: GPS/GNSS principles and receivers.	Ch.10–TB, RB4	CLO5
M6/Selected Topics	L40–L41: GPS/DGPS applications; L42–L43: Recent trends in SatCom	Research articles (e.g., IEEE ComSoc)	CLO5, CLO6

6. Evaluation Scheme:

Component	Wt (%)	Duration	Date & Time	Comments
Quizzes (two; one buffer, best of two counted)	15	40–45 min each	TBA in class	Closed book; formula sheet may be provided
RPBA* (max. two papers)	15	TBA	TBA	Open book
Classroom hands-on sessions / team discussions [#]	5	In class	Throughout semester	Linked to attendance; Hands-on sessions/discussions
Mid-Semester Test	25	90 min	9/10 (FN1)	Closed book; formula sheet(s) may be provided
Comprehensive Examination	40	180 min	13/12 (FN)	Closed book (CB) 16% + Open book (OB) 24%

* RPBA: Research Papers-Based Assignment (including viva) –

[#] Includes in-class problem-solving/hands-on sessions/simulation tasks etc.

All course learning outcomes are assessed through a combination of the evaluation components described above.

NC Criterion: A student should obtain 20% of the average of the class, or 10% of the course total marks, whichever is lower to clear the course. If any student gets marks lower than the prescribed standard mentioned above, he/she may be awarded NC.

Examination Policy—Electronic Gadgets Students should not keep any electronic gadget (such as mobile phones, headphones, etc.), excluding a calculator (if permitted), with them during any evaluation component — Quizzes, RPBA, Mid-Semester Test, Comprehensive Examination, and surprise in-class team tasks/discussions. If a student is found with such a gadget during any of these, they will be recommended to AUGSD for cancellation of course registration, or the scores obtained in that component will be declared void.

7. Attendance Policy

Each student is expected to attend all classes and participate in all evaluation components.

8. Office Consultation Hour:

Will be announced in the class. Also, feel free to email me to seek a prior appointment.

9. Make-up Policy:

No make-up will be given for the quizzes, in-class sessions/tasks. However, for the mid-semester and the comprehensive exam, makeup will be conducted for valid reasons with supporting documents and prior permission.

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EEE F472